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High Performance Ecoinformatics

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MSc Computer Science
Data Science and Engineering Curriculum

High Performance Econinformatics

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DEDICATION

Abstract Ecoinformatics is a modern method of doing science that enables scientists to generate new knowledge through innovative tools and approaches. Ecology is increasingly becoming a data-intensive science, involving the acquisition, preservation, and analysis of vast amounts of data from a variety of sources, including satellite and airborne remote sensing, instruments, sensors, and human observations. Ecologists are increasingly addressing larger-scale questions of both scientific and societal relevance, such as climate change and its consequences for ecosystems and related ecosystem services in the future. The topic of this thesis project stems from a research study conducted by the Department of Earth, Environmental, and Life Sciences (DISTAV, University of Genoa) that analyzes the shift in the distribution of natural habitats in the Alps due to future climate change to support decision-making processes regarding conservation planning and prioritization. The analysis is conducted using Species Distribution Models (SDMs), a dataset of approximately 3 million data points, and future climate change scenarios based on General Circulation Models (climate models). The statistical software R (<https://www.r-project.org/>) is used to generate future predictions of habitat distribution changes and biodiversity loss. The thesis candidate will work on improving the script and algorithmic functions, written in R, to streamline model analysis based on their knowledge and computer science background. This thesis project leverages interdisciplinary collaboration between DIBRIS and DISTAV to achieve better scientific results. Depending on the candidate's interest and quality of results, the thesis can be upgraded with asterisk.

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Chapter 1

Introduction and Motivation

introduzione deve essere scritta alla fine

1.1 Project context (taken from **Methods_GC_LD.pdf**)

Thesis proposal, ecoinformatics, Lucia.

Eco-informatics is a modern way of conducting scientific research that enables scientists to generate new knowledge through innovative tools and approaches. Ecology is increasingly becoming a data-intensive science that involves acquiring, storing, and analysing large amounts of data from a range of sources, including satellite and aerial remote sensing, instruments, sensors, and human observations. Ecologists increasingly address large-scale questions with both scientific and social relevance, such as climate change and its future effects on ecosystems and the ecosystem services associated with them. The topic of this thesis project originated from a research study conducted by the Department of Earth, Environmental and Life Sciences (DISTAV, University of Genoa). The study analyses changes in the distribution of natural habitats in the Alps caused by future climate change in order to support decision-making on conservation planning and prioritisation. The analysis uses Species Distribution Models, a dataset of approximately four million records, and future climate-change scenarios based on General Circulation Models. The statistical software R (<https://www.r-project.org/>) is used to generate predictions of future changes in habitat distribution and biodiversity loss. The thesis student will improve the working script and its algorithmic functions to make the model analysis more efficient, drawing on their knowledge and studies in computer science. This thesis project involves interdisciplinary collaboration between DIBRIS and DISTAV, which can help produce better scientific results.

New draft for supervisor review

The proposal text above reports approximately four million records. The current working dataset contains 2,583,359 records; confirm the final value with the supervisors.

1.2 Thesis motivation

The current workflow models 167 Alpine grassland species from 2,583,359 presence records. Its environmental rasters contain 63,951,097 cells and five variables. For each species, the production configuration calibrates five algorithms and builds two ensemble models. It projects these models onto the current environment and eight future climate scenarios. The workload combines a large spatial domain with repeated model calibration and raster projection.

The computational method determines whether the analysis can be completed at this scale. Historical runs lasted many hours. Several ended because of memory exhaustion, failed process forks, time limits, or file-writing collisions. Running the script on an HPC system does not solve these problems by itself. The software must control memory use, raster storage, and the scheduling of independent species.

This thesis examines the R workflow as HPC software. Leonardo provides the execution platform and the environment for measuring the program under different hardware allocations. CPU count, available RAM, storage policy, and filesystem activity affect runtime and completion. Our aim is to reduce execution time while preserving the scientific configuration, reproducibility, and output structure required by the ecological study.

1.3 Objectives and contribution

The scientific model and its parameters originate from the DISTAV study. This thesis contributes the software and experimental work required to execute that model at production scale. We adapted the R workflow to Leonardo and packaged its dependencies in a Singularity container. Each Slurm array task processes one species. We added runtime configuration, phase timings, resource measurements, success markers, and output checks. We also diagnosed memory, process-fork, and concurrent-write failures. We then compared worker counts and BIOMOD2 storage modes.

The first objective is to establish a reproducible baseline. Timings from runs with different inputs and parameters cannot provide such a baseline. The experiments identify the phases that dominate runtime. They also measure how worker count and storage settings affect elapsed time and memory. These measurements support later work on model parallelism

and raster I/O. The thesis focuses on software performance. The ecological validity of the model choices and evaluation metrics remains under review with the DISTAV collaborators.

1.4 Preliminary evaluation method

The evaluation separates controlled benchmarks from historical and diagnostic runs. Controlled comparisons keep the species, input data, random seed, and scientific parameters fixed. Each comparison changes one execution setting. For every run, we record wall time, phase timings, CPU use, MaxRSS, completion status, and output validation. Failed, cancelled, and incomplete jobs show the resource limits of a configuration. We retain them as evidence but do not use them as successful performance measurements.

End of new draft for supervisor review

Chapter 2

Background

real world and eventually related work

2.1 Ecoinformatics

2.2 Biological study context

An internal DISTAV methods document describes the biological study supported by the computational workflow. The study examines how climate change may alter the suitability of Alpine grassland habitats. It considers ten dry, alpine and subalpine habitat types from the EUNIS classification and 167 diagnostic plant species. Occurrence records are associated with environmental predictors at a spatial resolution of approximately 1 km. The predictors describe climate, soil and topographic heterogeneity. Species Distribution Models combine five modelling techniques and ensemble forecasts to estimate present and future habitat suitability.

The biological protocol also defines the analyses that follow model projection: changes in suitability, spatial shifts, habitat-level hotspot and cold-spot maps, and the relationship between suitability changes and species tolerance. This document provides the scientific context for the computing work, but it is not treated as the exact specification of every run. Dataset sizes, model repetitions and available climate scenarios changed during the project; the computational chapters therefore report the configuration recorded by the corresponding scripts and run manifests.

2.3 Data used

2.4 Related work

Chapter 3

Code as is

Existing workflow and computational problem.

3.1 R

3.2 BIOMOD2

3.3 Code pipeline

3.4 Original execution times

3.5 Improved execution times

3.6 Preliminary measurements on Spartaco

Before the Leonardo experiments, Lucia manually recorded four trial runs on Spartaco. The worksheet records per-phase time and RAM observations for two species, with different pseudo-absence settings and CPU counts. Table 3.1 preserves selected values from those records.

The measurements are useful for reconstructing the starting point of the project. They are not a controlled benchmark: the worksheet does not identify the code revision, container,

hardware configuration, input version or measurement method. The source label `n_cells` is retained in the table because its meaning cannot be established from the worksheet alone. These results are therefore not compared directly with the later Leonardo measurements.

Table 3.1: Selected fields from four manually recorded trial runs on Spartaco. `n_cells` retains the source label. Time units are those recorded in the original worksheet; the maximum RAM is the largest phase value in that row.

Species	<code>n_cells</code>	PA	Rep.	RUN	CPU	Formatting	Models	Future projection	Max RAM (GB)
<i>Omalotheca hoppeana</i>	681	10000	10	10	5	1.9 h	11.33 min	3.65 min	52.9
<i>Omalotheca hoppeana</i>	681	10000	10	10	10	2.02 h	7.74 min	8.07 min	56.16
<i>Agrostis capillaris</i>	100000	5000	5	5	5	32.25 min	53.87 min	3.73 min	26
<i>Agrostis capillaris</i>	100000	10000	5	5	5	59.37 min	1.6 h	4.22 min	29.96

Chapter 4

My new code

Design and implementation of the improvements.

4.1 Container

4.2 Parallel model execution

4.3 Parallel I/O

Chapter 5

Experimental results

Experimental methodology and results.

New draft for supervisor review

Convert the approved results to numbered LaTeX tables after supervisor review.

5.1 Leonardo

The experiments ran on Leonardo DCGP nodes. Each node provides 112 CPU cores and 494,000 MB of allocatable memory. A Slurm array task processed one species on one node. The full-raster runs used five environmental variables over 63,951,097 cells. They produced projections for the current environment and eight future scenarios. Most experiments requested all allocatable node memory. This reserved the whole node for accounting purposes, even when the R process used fewer CPU cores.

We evaluated completion separately from resource use. A zero process exit code was not sufficient when Slurm recorded an OOM event. A Slurm failure state also did not identify the failing R phase by itself. A run counted as complete only when its logs reached the end of the workflow. The expected timing files, output directories, non-empty files, and success marker also had to be present.

5.2 Tests

The first validated full-raster reference used four BIOMOD2 workers in F/F storage mode. Both projection storage options were disabled. The run completed in 22 hours, 33 minutes,

and 45 seconds. Slurm reported a MaxRSS of 260.94 GiB. The species directory contained 1,364 non-empty files and occupied 7.18 GiB. Four additional summary files were written at the output root. The outputs included 125 individual models, two ensemble models, and all eight future scenarios.

The F/T configurations used stacked projections and failed during the first future scenario. The two-worker run stopped after 5 hours and 44 minutes. The four-worker run stopped after 3 hours and 57 minutes. In both cases, `mcfork()` could not allocate memory. Their Slurm MaxRSS values were 413.22 and 413.80 GiB.

Increasing the worker count reduced elapsed time for the validated F/F configuration. Six workers completed in 17.85 hours, with a speedup of 1.264 over four workers. Eight workers completed in 15.09 hours, with a speedup of 1.495. Sixteen workers completed in 10 hours and 21 minutes, with a speedup of 2.179. The accounting, timing files, output structure, and success marker of the 16-worker run were validated.

Slurm MaxRSS was 258.95 GiB with six workers and 278.67 GiB with eight workers. It rose to 464.77 GiB with sixteen workers. CPU allocation efficiency fell from 68.1% to 60.0% and then 45.1%. A 32-worker F/F run reached 481.16 GiB. It was killed during the current projection after 1 hour and 40 minutes.

Future projections accounted for 77.4% to 83.5% of wall time in the validated runs. Formatting remained between 4,485 and 4,601 seconds. The current ensemble projection remained between 1,422 and 1,517 seconds. Neither phase showed a clear benefit from additional workers. The future loop fell from 67,819 seconds with four workers to 28,855 seconds with sixteen workers.

A separate controlled benchmark compared split and monolithic R implementations on the same node. Five runs of each version used a fixed seed, one worker, a small input, and no future projections. The split version averaged 26 minutes and 10 seconds with 47.09 GiB MaxRSS. The monolithic version averaged 26 minutes and 28 seconds with 47.63 GiB MaxRSS. The observed differences were 18 seconds and 0.54 GiB in favour of the split version. These results do not establish equivalence between the implementations. They also do not establish effects on parallelism or reproducibility.

The production configuration used ten pseudo-absence repetitions instead of five. It retained five cross-validation repetitions. Three species completed with this configuration. *Achillea atrata* took 25 hours, 36 minutes, and 36 seconds. *Achillea clusiana* took 21 hours, 26 minutes, and 42 seconds. *Agrostis capillaris* took 33 hours, 58 minutes, and 34 seconds. Their Slurm MaxRSS values were 316.16, 315.78, and 291.42 GiB, respectively.

Each species directory contained 2,614 non-empty files and a success marker. Four additional summaries were present for each run. The directories occupied 13.98, 13.36, and 59.13 GiB. In the recovered phase timings, the future loop ranged from 59,841 to 89,815

seconds. Individual modelling ranged from 172 to 10,989 seconds. This configuration performs more model runs than the earlier worker-count experiment, so their timings must not be compared directly.

5.3 Discussion and lessons learned

Increasing BIOMOD2 workers from four to sixteen reduced full-raster runtime in the validated F/F runs. Memory use rose sharply at sixteen workers, and the 32-worker run did not complete. The successful four-worker T/F run was 0.9% faster than the F/F reference and used 3.75 GiB more peak memory. Each mode was run once, and the exact code revision of the F/F reference is not recorded. The comparison therefore does not establish a storage effect. The F/T configurations, which enabled stacked projections, failed before the first future projection could complete.

Failed runs remain part of the evaluation. A failed run can reach its stopping point sooner when given more workers. This shorter time is not a speedup because the run produces no valid result. Runtime comparisons must include completion status, MaxRSS, and output checks. They must also keep scientific parameters fixed. The production runs with ten pseudo-absence repetitions answer a different performance question from the earlier runs with five.

Resource allocation affected both cost and runtime. The three completed production species consumed about 9,076 CINECA local hours. Each task reserved all allocatable node memory. The replacement array for species 4 to 167 did not start R. CINECA cancelled it in the administrative prolog after the project exceeded its budget. Future campaigns need a memory request based on measured MaxRSS, with a margin for variation between species. They should not request the full node unconditionally.

End of new draft for supervisor review

Chapter 6

Conclusions and future development

6.1 Conclusions

6.2 Future development

6.3 Ecoinformatics

What it is.

Why large datasets are used.

Why high-performance computers are used.

Purpose.

6.4 Data used

In the **Eval...** matrix, results with **calibration** below 0.5 are considered less performant.

6.5 Computer science objective

To be defined.

6.6 Interdisciplinary collaboration

Optional section.

Chapter 7

Appendix

Temporary or optional chapter.

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